FLYBOAT

Guiyang High-Tech Yige Electronic Co., Ltd.

34S Series Metal Oxide Varistors — Datasheet

Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- For top-grade 34S the melt-through point is confined within a 0.5 cm radius of the chip centre, converting a random failure location into a controlled one.
- If the chip does melt through, the flame jet is shielded by the copper electrode, which conducts the heat to the trip point and drives the thermal disconnector open.
- Exceptional damp-heat resistance: under a customer's severe boil test (30 min in boiling water, cool, measure, re-boil; 6 cycles required with no significant change in varistor voltage or leakage), Yige product has reached 8 cycles with small-current parameters essentially unchanged.
- High energy headroom: In 20 kA (25 kA for H variants), Imax up to 60 kA, 2 ms energy absorption up to 1,980 J.
- Long, verifiable service life (Arrhenius law): verified minimum expected life of 110 years at 40 °C, 11 years at 65 °C, 0.7 years at 95 °C upper ambient temperature (115 °C / 1,000 h ageing basis).
- High TOV withstand: at a temporary overvoltage of 1.45 × Uc the chip stays thermally stable for more than 45 min (r.m.s.), with no damage.

Model naming: 34S K 431 — 34 = nominal size (mm), S = square; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; /D = two-chip parallel, /T = three-chip parallel; -U = uncoated bare electrode disc (no mark = epoxy-coated by default).

1. Electrical characteristics — standard range

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms). Certification marks reproduce the catalog ratings table.

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
34SK101	100	±10	60	85	20	40	360-3.6	190	✓	✓	✓
34SK121	120	±10	75	100	20	40	400-3.6	230	✓	✓	✓
34SK151	150	±10	95	125	20	40	450-3.0	290	✓	✓	✓
34SK181	180	±10	115	150	20	40	490-2.8	340	✓	✓	✓
34SK201	200	±10	130	170	20	40	520-2.6	390	✓	✓	✓
34SK221	220	±10	140	180	20	40	550-2.6	440	✓	✓	✓
34SK241	240	±10	150	200	20	40	600-2.6	460	✓	✓	✓
34SK271	270	±10	175	225	20	40	680-2.6	530	✓	✓	✓

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
34SK301	300	±10	195	250	20	40	770-2.6	600	✓	✓	✓
34SK331	330	±10	210	270	20	40	830-2.55	650	✓	✓	✓
34SK361	360	±10	230	300	20	40	910-2.55	730	✓	✓	✓
34SK391	390	±10	250	320	20	40	1000-2.55	800	✓	✓	✓
34SK431	430	±10	275	350	20	50	1050-2.45	910	✓	✓	✓
34SK471	470	±10	300	385	20	50	1100-2.4	980	✓	✓	✓
34SK511	510	±10	320	410	20	50	1150-2.35	1050	✓	✓	✓
34SK561	560	±10	350	450	20	50	1250-2.35	1140	✓	✓	✓
34SK621	620	±10	385	505	20	50	1450-2.3	1240	✓	✓	✓
34SK621H	620	±10	385	505	25	60	1600-2.5	1280	—	—	—
34SK681	680	±10	420	560	20	50	1650-2.3	1250	✓	✓	✓
34SK681H	680	±10	420	560	25	60	1700-2.5	1280	—	—	—
34SK711	710	±10	440	585	20	40	1750-2.3	1260	✓	✓	✓
34SK751	750	±10	460	615	20	40	1800-2.3	1270	✓	✓	✓
34SK781	780	±10	485	640	20	40	1750-2.3	1280	✓	✓	✓
34SK821	820	±10	510	670	20	40	1800-2.3	1350	✓	✓	✓
34SK911	910	±10	550	745	20	40	2100-2.3	1460	✓	✓	✓
34SK951	950	±10	575	765	20	40	2200-2.3	1470	✓	✓	✓
34SK102	1000	±10	625	825	20	40	2400-2.3	1530	✓	✓	✓
34SK112	1100	±10	680	895	20	40	2700-2.3	1680	✓	✓	✓
34SK122	1200	±10	750	980	20	40	3000-2.3	1750	✓	✓	✓
34SK142	1400	±10	820	1120	20	40	3200-2.3	1780	✓	—	—
34SK152	1500	±10	850	1240	20	40	3400-2.3	1880	✓	—	—
34SK162	1600	±10	910	1320	20	40	3600-2.3	1920	✓	—	—
34SK182	1800	±10	1000	1465	20	40	4000-2.3	1980	✓	—	—

Notes: 34SK781's UP prints as 1750-2.3 in the catalog (identical to 34SK711) — transcribed as printed. The “H” variants (34SK621H / 34SK681H) are the high-throughput versions (In 25 kA, Imax 60 kA) and carry no certification marks in the catalog table.

2. Electrical characteristics — low-residual-voltage (low-clamping) range

Same column definitions as Table 1; optimised for lower clamping voltage at In.

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
34SK431	430	±10	275	350	20	40	950-2.2	910	✓	✓	✓
34SK471	470	±10	300	385	20	40	1030-2.2	980	✓	✓	✓
34SK511	510	±10	320	410	20	40	1130-2.2	1050	✓	✓	✓
34SK561	560	±10	350	450	20	40	1240-2.2	1140	✓	✓	✓
34SK621	620	±10	385	505	20	40	1350-2.15	1240	✓	✓	✓

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
34SK681	680	±10	420	560	20	40	1480-2.15	1250	✓	✓	✓
34SK711	710	±10	440	585	20	40	1540-2.15	1260	✓	✓	✓
34SK751	750	±10	460	615	20	40	1630-2.15	1270	✓	✓	✓
34SK781	780	±10	485	640	20	40	1670-2.15	1280	✓	✓	✓
34SK821	820	±10	510	670	20	40	1850-2.1	1350	✓	✓	✓
34SK911	910	±10	550	745	20	40	1940-2.1	1460	✓	✓	✓
34SK951	950	±10	575	765	20	40	2040-2.1	1470	✓	✓	✓
34SK102	1000	±10	625	825	20	40	2200-2.1	1530	✓	✓	✓
34SK112	1100	±10	680	895	20	40	2380-2.1	1680	✓	✓	✓
34SK122	1200	±10	750	980	20	40	2620-2.1	1750	✓	✓	✓

3. Electrical characteristics — high-energy range, 10/350 µs

Ratings are per single chip. Paralleling two or three chips achieves Iimp of 12.5 kA, 15 kA and 25 kA respectively. All rows: U1mA tolerance ±10 %. The catalog prints no UL / C-UL / TÜV marks for this table.

Model	U1mA (V)	UAC (V)	UDC (V)	In (kA)	Iimp (kA)	UP (V)	ETM (J)
34SK101	100	60	85	20	8	360-3.6	190
34SK121	120	75	100	20	8	400-3.6	230
34SK181	180	115	150	20	6	450-3.0	340
34SK201	200	130	170	20	6	490-2.8	390
34SK241	240	150	200	20	6	520-2.6	460
34SK271	270	175	225	20	6	550-2.6	530
34SK301	300	195	250	20	6	600-2.6	600
34SK331	330	210	270	20	6	680-2.6	650
34SK361	360	230	300	20	6	770-2.6	730
34SK391	390	250	320	20	6	830-2.55	800
34SK431	430	275	350	20	6	910-2.55	910
34SK471	470	300	385	20	6	1000-2.55	980
34SK511	510	320	410	20	6	1050-2.45	1050
34SK561	560	350	450	20	6	1100-2.4	1140
34SK621	620	385	505	20	6	1150-2.35	1240
34SK681	680	420	560	20	6	1250-2.35	1250
34SK711	710	440	585	20	6	1450-2.3	1260

4. Dimensions and thickness

Bare electrode disc (–U): square chip, face 33 ± 0.5 mm. Bare-disc leaded forms: 34SKaaaDEN-U, 34SKaaaOBO-U, 34SKaaaZP-U, 34SKaaaFG-U (aaa = voltage-code digits). Encapsulated (epoxy-coated) forms: TH, CQ, WK, G1, WEL, KL, LD, LD2, LWB, M, YL, PR; multi-chip encapsulated: HL/D, LD/D, LD2/D, LWB/D, M/D, KL/D, WK/D, YL/D, TH/T, ZK/T. Single-chip encapsulated body ≈ 35–36 mm square (verify individual dimensions against the outline drawings before tooling).

Disc thickness Tmax (mm), bare disc, standard & low-clamping ranges — part 1:

101	121	151	181	201	221	241	271	301	331	361	391	431	471	511	561	621	681
2.7	2.7	2.7	2.8	2.9	3.0	3.1	3.3	3.5	3.6	3.8	4.0	4.1	4.4	4.6	4.9	5.2	5.2

Disc thickness Tmax (mm), bare disc, standard & low-clamping ranges — part 2:

711	751	781	821	911	951	102	112	122
5.7	6.0	6.2	6.4	7.0	7.0	7.0	7.6	7.6

Disc thickness Tmax (mm), bare disc, high-energy (10/350) range — 101: 2.7 · 121: 2.7 · 151: 2.7 · 181: 3.3 · 201: 3.3 · 221: 4.2 · 241: 4.2 · 271: 4.9 · 301: 4.9 · 331: 5.5 · 361: 6.5 · 391: 6.5 · 431: 6.5 · 471: 8.0 · 511: 8.0 · 561: 9.0 · 621: 9.0 · 681: 9.0 · 711: 9.0..

Encapsulated thickness Tmax (mm), standard & low-clamping ranges — 101: 3.8 · 121: 3.8 · 151: 3.8 · 181: 3.9 · 201: 4.0 · 221: 4.1 · 241: 4.2 · 271: 4.4 · 301: 4.6 · 331: 4.7 · 361: 4.9 · 391: 5.1 · 431: 5.2 · 471: 5.5 · 511: 5.7 · 561: 6.0 · 621: 6.3 · 681: 6.3 · 711: 6.6 · 751: 7.1 · 781: 7.3 · 821: 7.5 · 911: 8.1 · 951: 8.1 · 102: 8.1 · 112: 8.9 · 122: 8.9 · 142: 9.9 · 152: 10.4 · 162: 10.9 · 182: 11.9..

Encapsulated thickness Tmax (mm), high-energy range — 431: 7.5 · 471: 7.5 · 511: 9.0 · 561: 9.0 · 621: 10.0 · 681: 10.0 · 711: 10.0..

Multi-chip bare-disc thickness Tmax (mm) — /D (two-chip): 431: 7.3 · 471: 7.9 · 511: 8.3 · 561: 8.9 · 621: 9.5 · 681: 9.5 · 711: 10.5 · 751: 11.1 · 781: 11.5 · 821: 11.9 · 911: 12.8 · 951: 13.1 · 102: 13.1 · 112: 14.3 · 122: 14.3.. /T (three-chip): 431: 11.4 · 471: 12.0 · 511: 12.4 · 561: 13.0 · 621: 13.6 · 681: 13.6 · 711: 14.6 · 751: 15.1..

5. General specifications

- Operating temperature: −40 °C to +85 °C (no derating required); storage −40 °C to +125 °C (unloaded).
- Leakage current: ≤ 10 µA stable at 75 % of U1mA (DC).
- Ageing: after 1,000 h at 125 °C at 100 % DC load, ΔU1mA within −10 %, leakage ≤ 100 µA, no visible mechanical damage.
- Damp heat: 85 °C / 85 % RH, 2,000 h at UAC — ΔU1mA within −10 %, leakage ≤ 60 µA.
- RoHS: compliant with EU Directive 2011/65/EU. Solderability: lead-out terminals uniformly tinned.
- Marking/packaging: carton packing with packing list, outgoing inspection report and certificate of conformity; custom labels on request.

Certification marks in this datasheet reproduce the catalog's own checkmark columns (standard table: 34SK101–34SK122 UL / C-UL / TÜV; 34SK142–34SK182 UL only; 34SK621H / 34SK681H unmarked; low-clamping table: all rows UL / C-UL / TÜV; high-energy table: unmarked). Source: company product catalog, transcribed 2026-10-19; verify against the current UL file listing before external release.